WEEK 8: ASSIGNMENT

1. Which of the following statement/statements are true for phase shifted full bridge converter?

(A) The leading leg switches achieves ZVS turn on even at the light load condition.
(B) The lagging leg switches achieves ZVS turn on even at the light load condition.
(C) The increased value of resonant inductor helps in improving the possibilities of lagging leg switches achieves ZVS turn on at light loading condition.
(D) Both the leading and lagging leg switches achieves ZVS irrespective of loading condition.

Answer: A, C

2. Which of the following statement/statements is/are correct for Phase shifted full bridge, Symbols have obvious meaning.
A. For the lagging leg switches, during turn-on the switch capacitive energy is removed using L_{lkg}
B. For the lagging leg switches, during turn-on the switch capacitive energy is removed using $L_{lkg} + \frac{L}{n^2}$
C. For the leading leg switches, during turn-on the switch capacitive energy is removed using L_{lkg}
D. For the leading leg switches, during turn-on the switch capacitive energy is removed using $L_{lkg} + \frac{L}{n^2}$

Answer: A, D

3. The phase shifted full bridge converter is driven from input voltage of 200V, 1:1 transformer, obtains the output voltage of 100V and delivers 500W load. The leakage inductance of the transformer is 40µH. The switches used have the output capacitance of 100 pF, then what will be the minimum deadtime (ns) to be provided for leading leg switches to achieve the zero voltage turn-on of them. Assume, the current ripple in the filter inductor is negligible. The answer should be rounded up to 2 decimal places.

Answer: 8

Range can be given from 6 to 10

4. The phase shifted full bridge converter with input voltage of 100V, 1:1 transformer, delivers to 10 Ω load at the output voltage of 50V. The leakage inductance of the transformer is 10µH. The switches have the output capacitance of 500 pF, then what will be the minimum deadtime (ns) to be provided for lagging leg to achieve the soft turn on of the lagging leg. Assume the

output inductor current ripple is negligible. The answer should be rounded up to 2 decimal places and 1 $radian = 57.3\ degrees$.

Answer: 100.369

Range can be given from 90 to 110

5. A phase shifted full bridge converter is driven from the input voltage of 200V. The converter feeds a 5A load and generates an output voltage of 95V. If the allowable duty loss is 5% and switching frequency is kept at 50 kHz, what will be the resonant inductor value in µH. Assume the resonant inductor is obtained from the leakage inductance of the transformer and transformer has 1:1 turns. The answer should be rounded up to 2 decimal places.

Answer: 10

Range can be given from 8 to 12

6. If the ideal full bridge converter is operated with phase shift modulation has 180 degree phase shift between the diagonal switches of the full-bridges. Then what will be the obtained output voltage? Neglect the effect of duty loss, switch drop and output voltage ripple is negligible.

A. 0
B. Equal to input voltage
C. Greater than the input voltage
D. Equal to half the input voltage

Answer: A

7. For the phase shifted full bridge converter having 1:1 transformer, switches with same capacitance and negligible ripple in load current. Which of the following statement/statements are true. Symbols have obvious meaning and I is the transformer primary winding current at the time of switching.

(A) The necessary condition to achieve ZVS of turn on of lagging leg switches is $\frac{1}{2}\left(L_{lkg}I^2\right) > CV_{in}^2$.

(B) The necessary condition to achieve ZVS of turn on of lagging leg switches is $\frac{1}{2}\left((L_{lkg} + \frac{L}{n^2})I^2\right) > CV_{in}^2$.

(C) The necessary condition to achieve ZVS of turn on of leading leg switches is $\frac{1}{2}\left((L_{lkg} + \frac{L}{n^2})I^2\right) > CV_{in}^2$.

(D) The necessary condition to achieve ZVS of turn on of leading leg switches is $\frac{1}{2}\left(L_{lkg}I^2\right) > CV_{in}^2$

Answer: A, C

8. The phase shifted full bridge converter has transformer with leakage inductance of 5 µH and primary winding self-inductance of 1000 µH. The output filter inductance used is 100 µH. If the switches used have the output capacitance of 100 nF, and the resonance frequency for the switches in lagging leg is required to be 100 kHz. Then, what will be the additional inductor (in µH) be place in series with the transformer primary winding to meet the required resonance frequency. The answer should be rounded up to 2 decimal places and $\pi = 3.14$

Answer: 20.356

Range can be given as 17 to 23

9. Which of the following statement/statements are true about the phase shifted full bridge converter?
A. The effect of $\widetilde{d_v}$ term due to change in input voltage have opposite effect on the effective duty ratio ($\tilde{d}$)
B. The effect of $\widetilde{d_v}$ term due to change in input voltage have same effect on the effective duty ratio ($\tilde{d}$)
C. The effect of $\widetilde{d_\iota}$ term due to change in inductor current have opposite effect on the effective duty ratio ($\tilde{d}$)
D. The effect of $\widetilde{d_\iota}$ term due to change in inductor current have same effect on the effective duty ratio ($\tilde{d}$)

Answer: B, C

10. If the ideal phase shifted full bridge converter has 72 degrees phase shift between the diagonal switches of the full-bridges, then what will be the ratio of output voltage to input voltage. It uses a 1:1 transformer. Neglect the effect of duty loss, and output voltage ripple is negligible. Note: The answer should be rounded up to 2 decimal places.

Answer: 0.6

Range can be given from 0.5 to 0.7

Answer Key:

1. A, C	2. A, D
3. 6 to 10	4. 90 to 110
5. 8 to 12	6. A
7. A, C	8. 17 to 23
9. B, C	10. 0.5 to 0.7